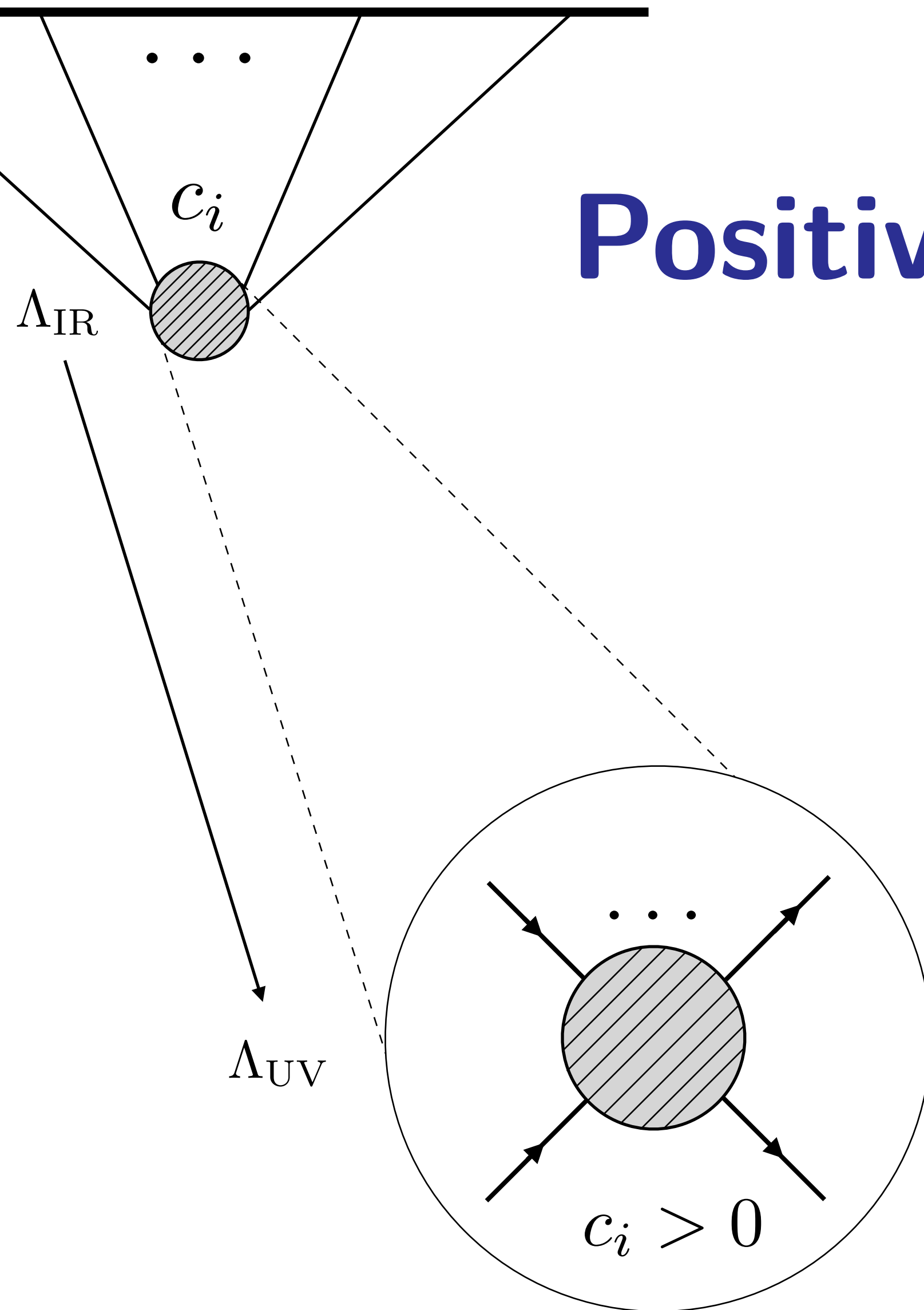


Positivity constraints on the EFT of inflation

Rafael Bravo

Pontificia Universidad Católica de Chile

Work in progress with S. Cespedes (UCHile), N. Parra (KIT) and S. Riquelme (USS)



Motivation:

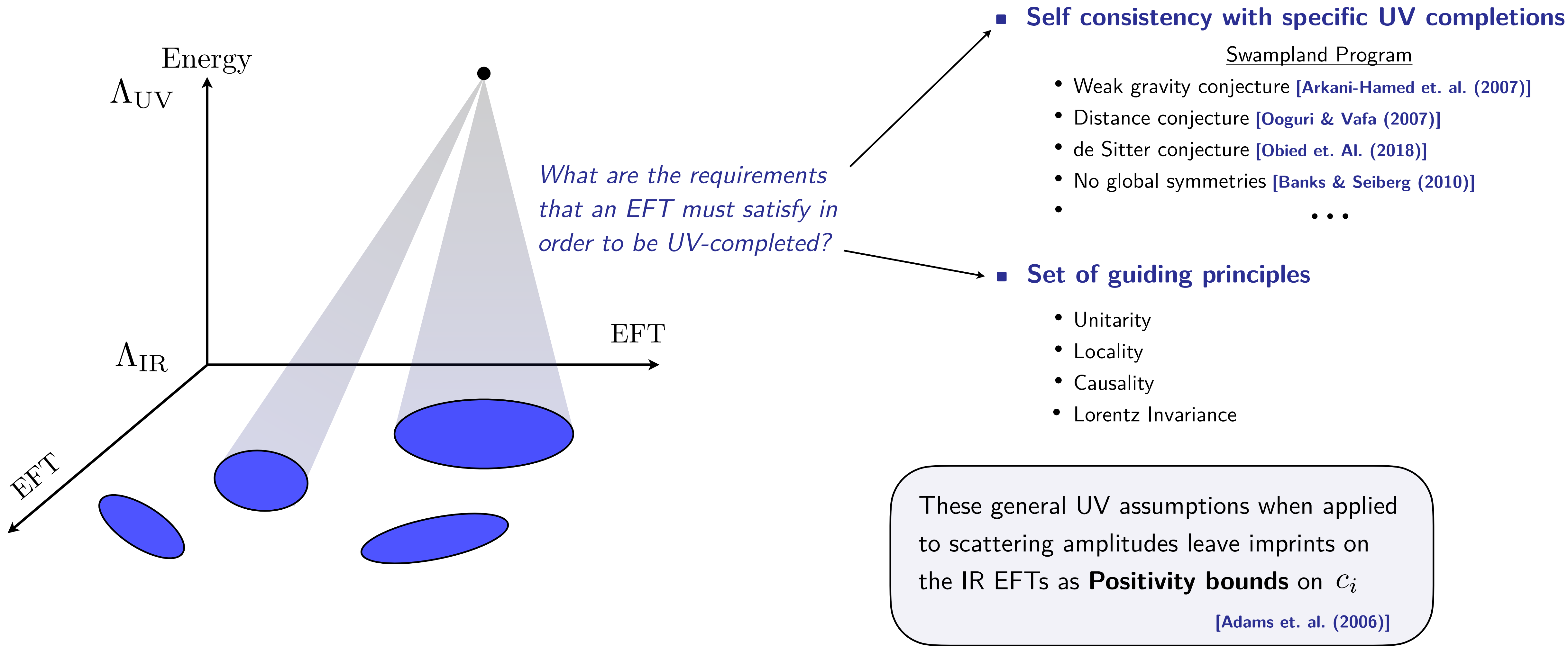
UV constraints on IR Physics

UV constraints on IR Physics

$$\mathcal{L}_{\text{EFT}}[\phi] = \mathcal{L}_0[\phi] + \sum_i c_i \frac{\mathcal{O}_i[\phi]}{\Lambda^{d_i-4}}$$

Q: Can every EFT be UV completed ?

A: No! Some low-energy EFT do not admit well defined UV completions



Positivity Bounds: a concrete example

- Low energy EFT of a massless shift symmetric scalar

$$\mathcal{L} = -\frac{1}{2}\partial_\mu\pi\partial^\mu\pi + \frac{c_1}{\Lambda^4}(\partial_\mu\pi\partial^\mu\pi)^2 \quad \pi \rightarrow \pi + c, \quad \pi \rightarrow -\pi$$

Expanding around a constant background gradient background $\partial_\mu\pi_0 = C_\mu$ $\pi = \pi_0 + \delta\pi$

$$k^\mu k_\mu + 4\frac{c_1}{\Lambda^4}(C \cdot k)^2 = 0 \quad (C \cdot k)^2 > 0 \quad \text{Subluminality} \implies c_1 > 0$$

Causality $\implies$ positivity

- If we were started from the UV theory of a broken U(1)

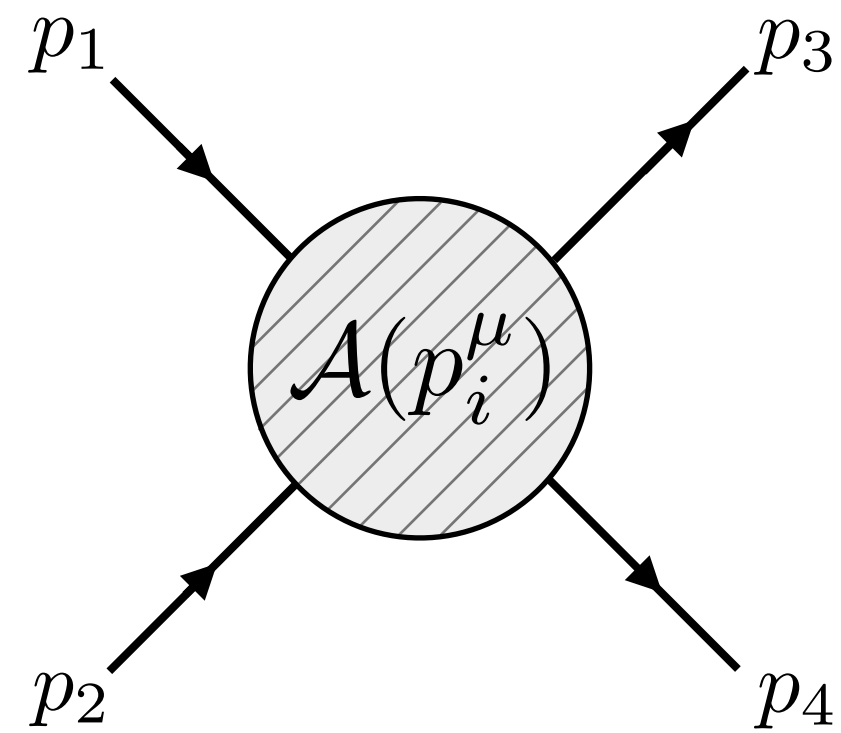
$$\mathcal{L} = -\frac{1}{2}(1+h)(\partial\pi)^2 - \frac{1}{2}(\partial h)^2 - M^2 h^2 - \frac{\lambda}{4}(h^2 + 2vh)^2$$

Integrating out h at tree-level

$$\mathcal{L}_{\text{EFT}}[\pi] = -\frac{1}{2}\partial_\mu\pi\partial^\mu\pi + \frac{\lambda}{M^4}(\partial_\mu\pi\partial^\mu\pi)^2 \quad \lambda > 0$$

Positivity Bounds from scattering amplitudes in flat space

$2 \rightarrow 2$ scattering for identical massive scalars of mass m



$$s + t + u = 4m^2$$

Properties

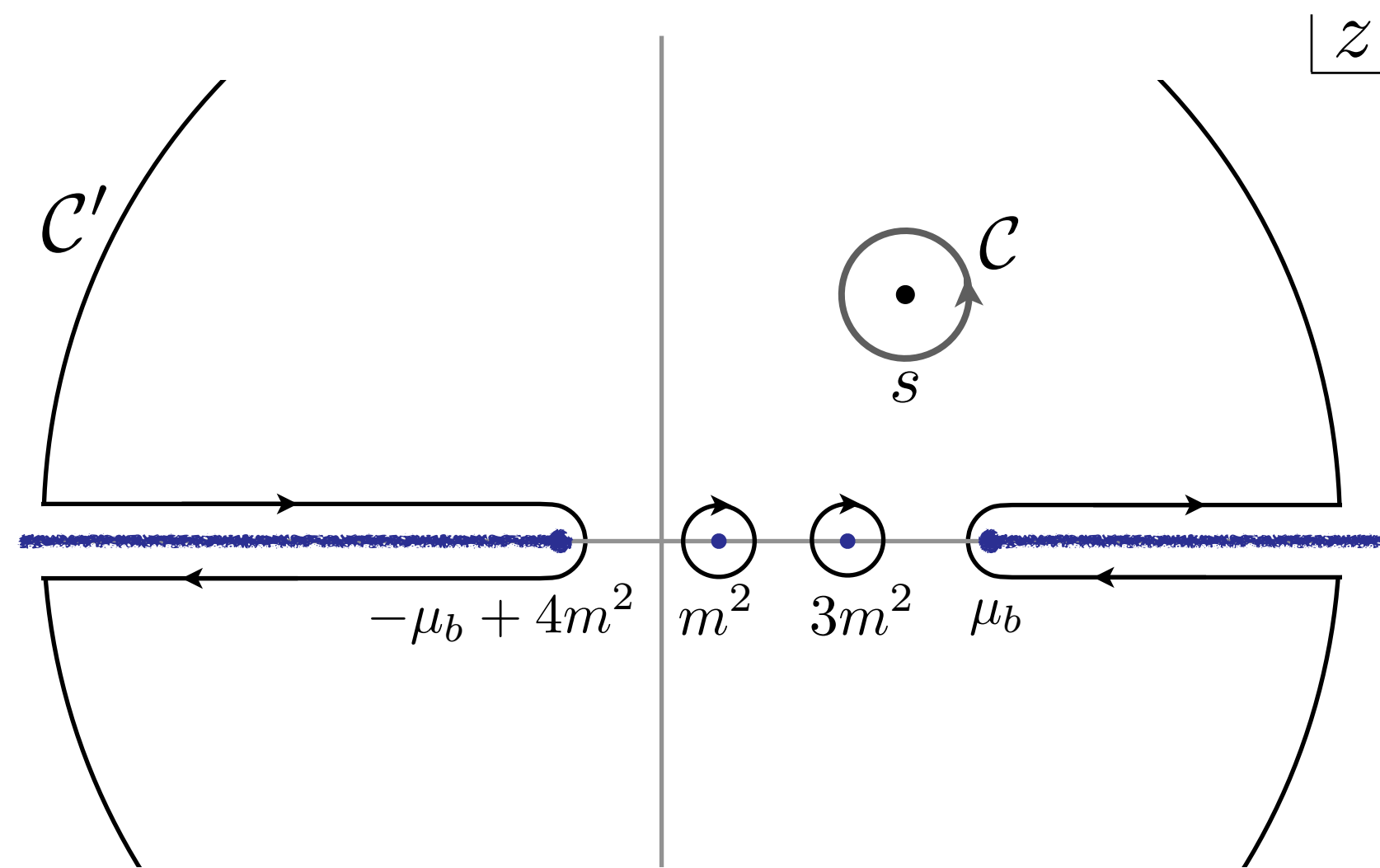
- Unitarity: $S^\dagger S = 1 \implies \text{Im } \mathcal{A}(s, 0) \propto s \sigma_{\text{tot}}(s) \geq 0$
- Locality: $|\mathcal{A}| \leq C s \log^2 s$
- Causality: $\mathcal{A}(s, t)$ is analytic
- Lorentz invariance $s \leftrightarrow -s$

Positivity Bounds

$$\left. \frac{\partial^{2n} \mathcal{A}(s)}{\partial s^{2n}} \right|_{s=0} \geq 0$$

[Adams et. Al (2006)]

[Arkani-Hamed, Huang, Huang (2020)]



$$\mathcal{A}(s, t) = \frac{1}{2\pi i} \oint_C dz \frac{\mathcal{A}(z, t)}{z - s}$$

We can expand the amplitude in the forward limit $t \rightarrow 0$ as

$$\mathcal{A}(s, 0) = \mathcal{A}(0, 0) + s \mathcal{A}'(s, 0) + \frac{s^2}{\pi} \int_{4m^2}^{\infty} \frac{\text{Im} \mathcal{A}(z, 0)}{z^2(z - s)} dz + \dots$$

$$\mathcal{A}''(0, 0) = \frac{2}{\pi} \int_{4m^2}^{\infty} dz \frac{\text{Im} \mathcal{A}(z, 0)}{z^3} = \frac{2}{\pi} \int_{4m^2}^{\infty} dz \frac{\sigma_{\text{tot}}(z)}{z^2} \geq 0$$

For the theory

$$\mathcal{L} = -\frac{1}{2} \partial_\mu \pi \partial^\mu \pi + \frac{c_1}{\Lambda^4} (\partial_\mu \pi \partial^\mu \pi)^2$$

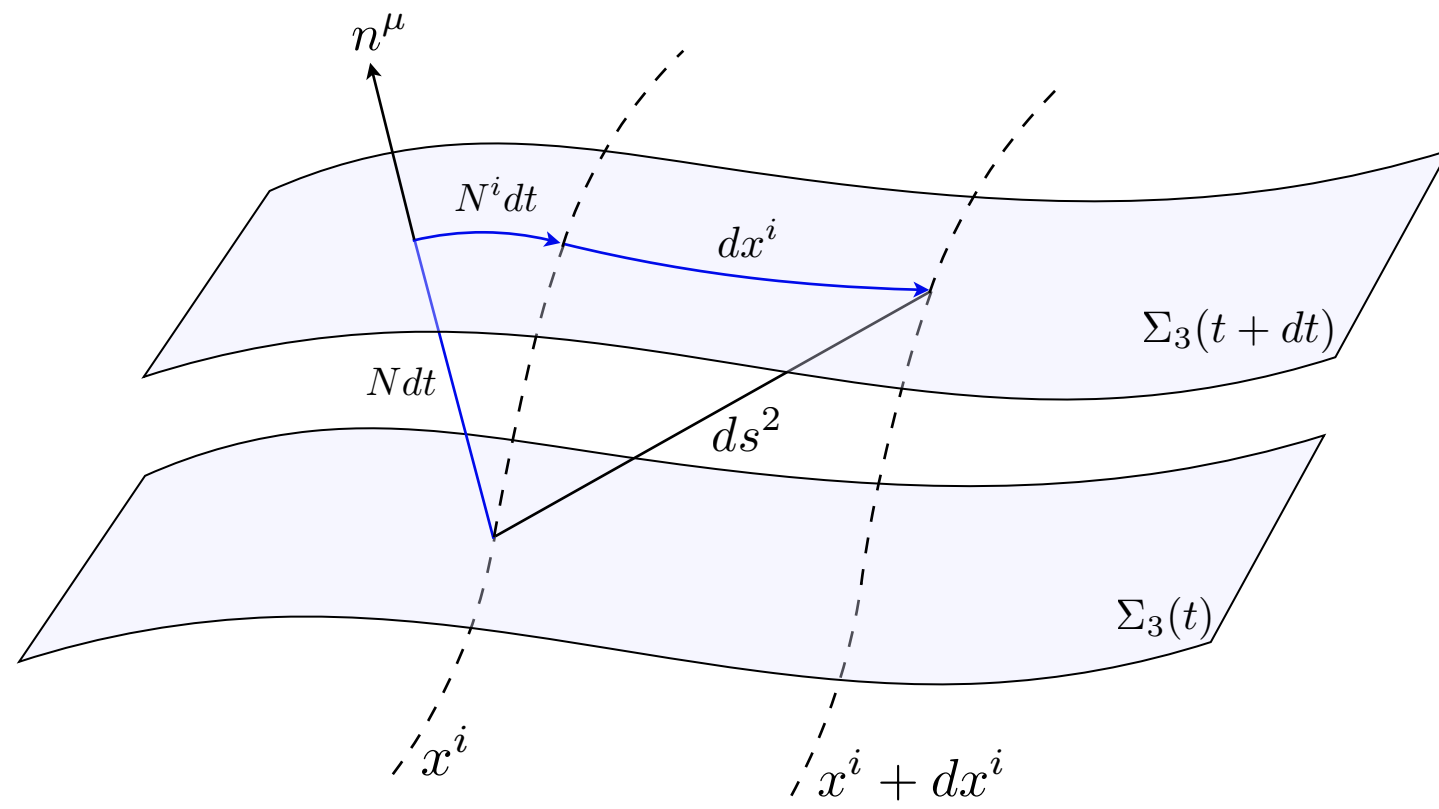
The corresponding 2-2 scattering amplitude

$$\mathcal{A}(s, 0) = 2 \frac{c_1}{\Lambda^4} s^2$$

The positivity bound implies $c_1 \geq 0$

Analytic structure in the forward limit $\mathcal{A}(s, t = 0)$

The EFT of inflation



Unitary gauge action

$$S = \int d^4x \sqrt{-g} \left[\frac{M_{\text{Pl}}^2}{2} R + M_{\text{Pl}}^2 \dot{H} g^{00} - M_{\text{Pl}}^2 (3H^2 + \dot{H}) + \frac{M_2^4(t)}{2!} (\delta g^{00})^2 + \frac{M_3^4(t)}{3!} (\delta g^{00})^3 \right. \\ \left. - \frac{\bar{M}_1^3(t)}{2} (\delta g^{00}) \delta K^\mu{}_\mu - \frac{\bar{M}_2^2(t)}{2} (\delta K^\mu{}_\mu)^2 - \frac{\bar{M}_3^2(t)}{2} \delta K^\mu{}_\nu \delta K^\nu{}_\mu + \dots \right]$$

[Cheung et. al (2007)]

$t \rightarrow t + \pi(t, x)$ decoupling limit Stückelberg action

$$S_\pi = \int d^4x \left[M_{\text{Pl}}^2 \dot{H} (\partial_\mu \pi)^2 + 2M_2^4 \left(\dot{\pi}^2 + \dot{\pi}^3 - \dot{\pi} \frac{1}{a^2} (\partial_i \pi)^2 \right) - \frac{4}{3} M_3^4 \dot{\pi}^3 - \frac{\bar{M}^2}{2} \frac{1}{a^4} (\partial_i^2 \pi)^2 + \dots \right]$$

Can we say something about the signs of $M_i, \bar{M}_i$?

[Baumann et. al. (2015)]

[Grall & S. Melville (2021)]

[Creminelli et. al. (2023)]

[Freytsis et. al. (2023)]

[Green et. al. (2023)]

[Serra & Trombetta (2024)]

[Creminelli et. al. (2024)]

[Gordon Lee & S. Melville (2025)]

■ ~~Flat space~~ $\Rightarrow$ quasi dS

■ ~~Scattering amplitudes~~ $\Rightarrow$ equal-time correlation functions

■ ~~Lorentz invariance~~ $\Rightarrow$ spontaneously broken, bonus: no energy conservation

Let's try with a spherical cow

Lorentz breaking amplitudes
and
Amplitudes from dS correlators

Scattering amplitudes in time-dependent backgrounds: superfluid phonon

$$S[\Phi] = \int d^4x \left(-\partial_\mu \Phi^\dagger \partial^\mu \Phi + M^2 \Phi^\dagger \Phi - \lambda (\Phi^\dagger \Phi)^2 \right) \quad O(2) \simeq U(1) \rtimes \mathbb{Z}_2 \quad \begin{array}{l} U(1) : \quad \Phi \rightarrow \Phi e^{i\alpha} \\ \mathbb{Z}_2 : \quad \Phi \rightarrow \Phi^\dagger \end{array}$$

Superfluid: spontaneously broken time-translations P_0 and the internal $U(1)$. Ground state invariant under $P_0 + \mu Q$

$$\Phi(x) = \frac{\rho(x)}{\sqrt{2}} e^{i\theta(x)} \quad \Longrightarrow \quad \mathcal{L}[\rho, \theta] = -\frac{1}{2}(\partial\rho)^2 - \frac{1}{2}(\partial\theta)^2 \rho^2 + \frac{M^2}{2}\rho^2 - \frac{\lambda}{4}\rho^4 \quad \begin{array}{l} U(1) : \quad \theta \rightarrow \theta + \alpha, \rho \rightarrow \rho \\ \mathbb{Z}_2 : \quad \theta \rightarrow -\theta, \rho \rightarrow \rho \end{array}$$

Superfluid phase: $\theta = \mu t + \pi$, $\rho = v + h$ with $v = \sqrt{\frac{\mu^2 + M^2}{\lambda}}$

$$\mathcal{L}[h, \pi] = -\frac{1}{2}(\partial_\mu h)^2 - \frac{v^2}{2}(\partial_\mu \pi)^2 + \frac{1}{2} [(\partial_\mu \pi)^2 + 2\mu\dot{\pi}] (h^2 + 2vh) - \frac{\lambda}{4}(h^2 + 2vh)^2$$

$$\begin{array}{l} \pi \rightarrow \pi + \alpha \\ \pi \rightarrow -\pi \quad t \rightarrow -t \end{array}$$

$$\omega_\pm^2(\mathbf{k}) = \mathbf{k}^2 + M^2 + 3\mu^2 \pm \sqrt{4\mathbf{k}^2\mu^2 + (M^2 + 3\mu^2)^2}$$

Gapless mode ω_- : phonon

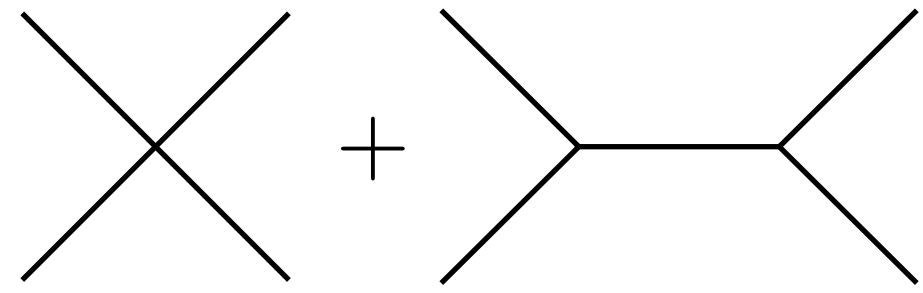
$\mu \rightarrow 0$ Recovers the Lorentz invariance

[Son (2002)]
[Son & Wingate (2006)]
[Berezhiani (2020)]

Tree level scattering amplitude

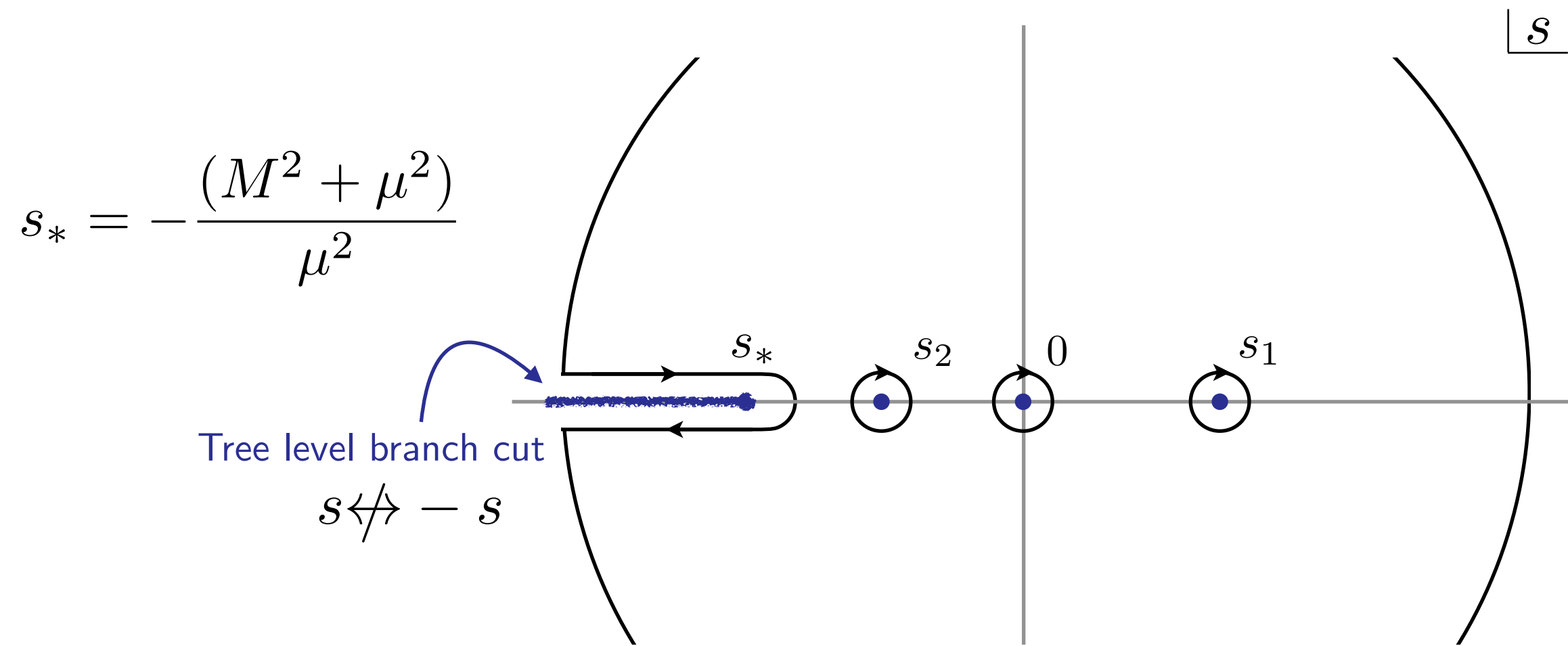
The phonon $\pi\pi \rightarrow \pi\pi$ scattering amplitude at tree level is found to be

[Creminelli et. al. (2023)]
[Hui et. al. (2023)]



$$\mathcal{A}(s, t, u) = \lambda \left(1 + \frac{1 - c_s^2}{c_s^2 \sqrt{1 + y}} \right)^{-2} (\mathcal{A}_c + \mathcal{A}_s + \mathcal{A}_t + \mathcal{A}_u)$$

$$y \equiv \frac{\mu^2 s}{(M^2 + \mu^2)^2}$$



$$s + t + u = M^2 + \mu^2 + \sqrt{\mu^2 s + (M^2 + \mu^2)^2}$$

In the center of mass and equal energies configuration

Low energy expansion in the forward limit

$$\mathcal{A}(s, 0) = \lambda \frac{2M^6 + 6\mu^2 M^4 + 9\mu^4 M^2 + 9\mu^6}{2(M^2 + \mu^2)^2 (M^2 + 3\mu^2)^3} s^2 - \lambda \frac{\mu^2 (12M^8 + 58M^6 \mu^2 + 117M^4 \mu^4 + 126M^2 \mu^6 + 63\mu^8)}{4(M^2 + \mu^2)^4 (M^2 + 3\mu^2)^4} s^3$$

Positive

Negative

Positivity (negativity) properties of the phonon amplitude

- Lorentz invariant case $\mu = 0$

$$\mathcal{A}(s)|_{\mu=0} = a_2 s^2 + a_3 s^3 + a_4 s^4 + \dots = \sum_{n=2}^{\infty} a_n s^n$$

$$s \rightarrow -s \implies a_{2m+1} = 0 \quad m \geq 0$$

$$a_{2m} > 0$$

- Superfluid phase $\mu \neq 0$

$$\mathcal{A}(s) = a_2(\mu) s^2 + a_3(\mu) s^3 + a_4(\mu) s^4 + \dots = \sum_{n=2}^{\infty} a_n(\mu) s^n$$

For arbitrary values of μ

$$a_{2m}(\mu) > 0, \quad a_{2m+1}(\mu) < 0$$

Generalized positivity condition

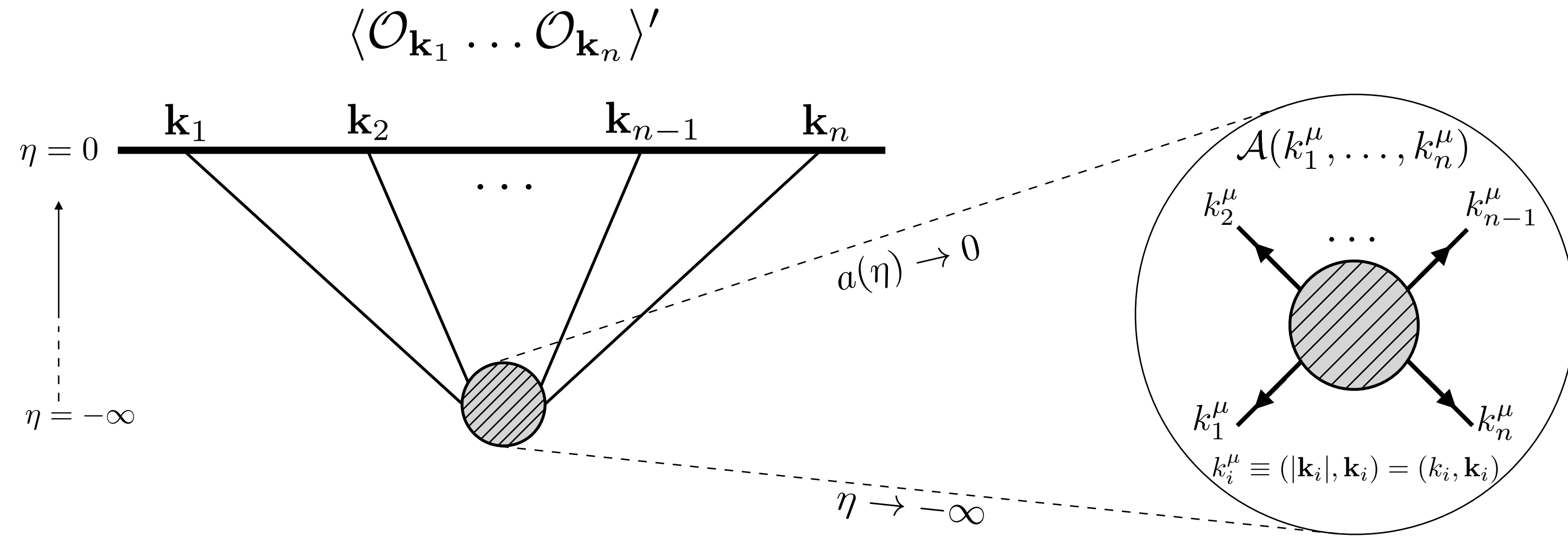
$$(-1)^n \frac{\partial^n \mathcal{A}}{\partial s^n} \Big|_{s=0} > 0$$

Could this be a general feature of spontaneously broken Lorentz theories?

Let's take it as a conjecture

The amplitude limit of cosmological correlators

Cosmological correlators contain flat scattering amplitudes!



$$\lim_{k_T \rightarrow 0} \langle \mathcal{O}_{\mathbf{k}_1} \dots \mathcal{O}_{\mathbf{k}_n} \rangle' = \frac{(-1)^n H^{p+n-1} \Gamma(p)}{2^{n-1} (\prod_{a=1}^n k_a^2)} \times \frac{\text{Re} (i^{1+n+p} \mathcal{A}(k_1^\mu, \dots, k_n^\mu))}{k_T^p}$$

Annotations:

- $k_T = \sum_a k_a$ (Total energy)
- $p = 1 + \sum_i (d_i - 4)$ (Mass dimension of the operator)
- k_T^p ("Energy" pole)

[Maldacena & Pimentel (2011)]
 [Arkani-Hamed & Maldacena (2015)]
 [Arkani-Hamed, Benincasa & Postnikov (2017)]
 [Pajer (2020)]

■ Scattering Amplitudes

$$\mathcal{A} \propto \int_{-\infty}^{\infty} dt \prod_a e^{e E_a t} = 2\pi \delta(E_T)$$

■ Correlation functions

$$\langle \mathcal{O}_{\vec{k}_1} \dots \mathcal{O}_{\vec{k}_n} \rangle' \propto \int_{-\infty}^0 d\eta \eta^{p-1} \prod_a e^{i k_a \eta} = \frac{1}{k_T^p}$$

At early times

$$a(\eta) \rightarrow 0$$

$$\varphi_k(\eta) \sim \frac{1}{\sqrt{2k}} e^{-ik\eta}$$

The mode functions are the same that in Minkowski

Positivity constraints on the EFT of inflation

Positivity constraints on the EFT of inflation

Let's compute the trispectrum in the EFT of inflation, extract the flat-amplitude and apply the generalized positivity condition coming from the superfluid

- First, let us consider the contributions from

$$S[g^{00}] = \int d^4x \sqrt{-g} \left[M_{\text{Pl}}^2 \dot{H} g^{00} - M_{\text{Pl}}^2 (3H^2 + \dot{H}) + \sum_{n \geq 2} \frac{M_n^4}{n!} (g^{00} + 1)^n \right]$$

$P(X)/\mathbf{K}$ - Inflation
DBI
...

Restoring time diffs $t \rightarrow t + \pi(t, x)$

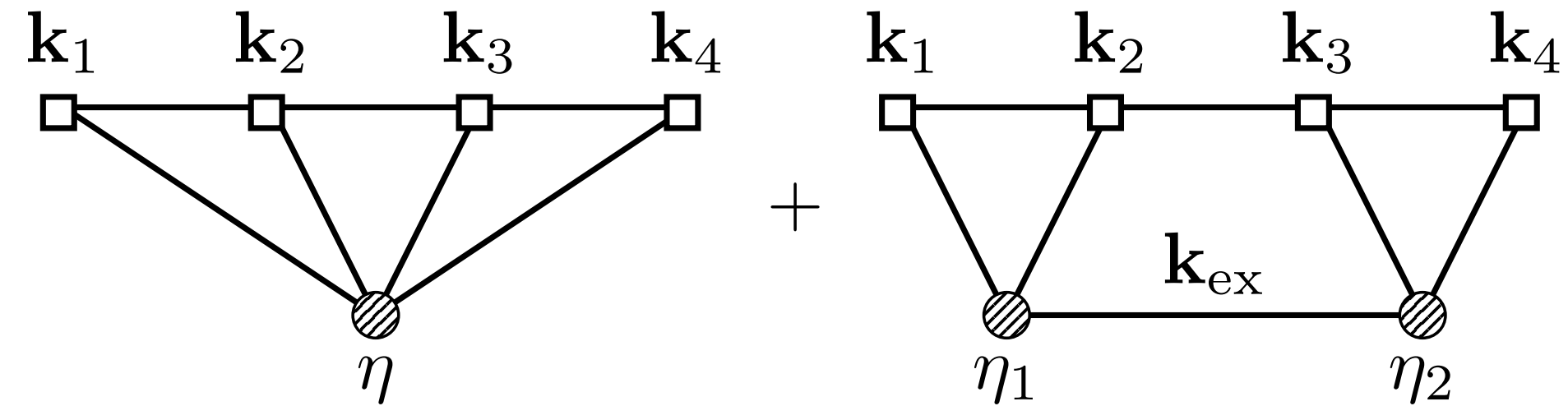
$$(g^{00} + 1) \rightarrow -2\pi + \dot{\pi}^2 + \frac{(\nabla \pi)^2}{a^2}$$

$\pi \rightarrow \pi + \alpha$ same as the
 $\pi \rightarrow -\pi \quad t \rightarrow -t$ superfluid

The action up to quartic order is

$$S[\pi_c] = \int d\eta d^3\mathbf{x} \left[\frac{a^2(\eta)}{2} (\pi_c'^2 - c_s^2 (\nabla \pi_c)^2) + \frac{a(\eta)}{\Lambda_\pi^2} (\alpha_1 \pi_c'^3 - \alpha_2 \pi_c' (\nabla \pi_c)^2) \right. \\ \left. + \frac{1}{\Lambda_\pi^4} (\beta_1 \pi_c'^4 - \beta_2 \pi_c'^2 (\nabla \pi_c)^2 + \beta_3 (\nabla \pi_c)^4) \right]$$

Positivity constraints on the EFT of inflation

$$\langle \hat{\pi}_c(\mathbf{k}_1) \hat{\pi}_c(\mathbf{k}_2) \hat{\pi}_c(\mathbf{k}_3) \hat{\pi}_c(\mathbf{k}_4) \rangle' =$$


The flat-space limit

$$\lim_{k_T \rightarrow 0} \langle \hat{\pi}_c(\mathbf{k}_1) \hat{\pi}_c(\mathbf{k}_2) \hat{\pi}_c(\mathbf{k}_3) \hat{\pi}_c(\mathbf{k}_4) \rangle' = -\frac{H^8 4!}{2^3 \prod_{a=1}^4 k_a^2} \frac{1}{k_T^5} \times \text{Re} [\mathcal{A}(k_a, \mathbf{k}_a)]$$

We extracted the following Amplitude

$$\mathcal{A}(s, t \rightarrow 0) = \left[\frac{3}{2}\beta_1 + \frac{\beta_2}{2} + \frac{3}{2}\beta_3 - \frac{9}{4}\alpha_1^2 - \frac{3}{4}\alpha_1\alpha_2 - \frac{\alpha_2^2}{4} \right] \frac{s^2}{\Lambda_\pi^4}.$$

The usual positivity bound $\mathcal{A}''(s) \big|_{s=0} > 0$

$$6\beta_1 + 2\beta_2 + 6\beta_3 > 9\alpha_1^2 + 3\alpha_1\alpha_2 + \alpha_2^2$$

Agrees exactly with the result of [\[Baumann et. al. \(2015\)\]](#)

$$\tau_{\text{NL}} \geq \left(\frac{6}{5} f_{\text{NL}} \right)^2$$

[\[Suyama & Yamaguchi \(2007\)\]](#)

The trispectrum is dominated by the contact interaction

Positivity constraints on the EFT of inflation

s^3 terms are produced by higher-derivative operators, i.e., those generated by the extrinsic curvature $\delta K^\mu{}_\nu$

$$S[\delta K^\mu{}_\nu] = \int d^4x \sqrt{-g} \left[-\frac{\bar{M}_5^2(t)}{3!} (\delta g^{00}) (\delta K^\mu{}_\mu)^2 - \frac{\bar{M}_6^2(t)}{3!} (\delta g^{00}) \delta K^\mu{}_\nu \delta K^\nu{}_\mu \right. \\ \left. - \frac{\bar{M}_{11}^2(t)}{4!} (\delta g^{00})^2 (\delta K^\mu{}_\mu)^2 - \frac{\bar{M}_{12}^2(t)}{4!} (\delta g^{00})^2 \delta K^\mu{}_\nu \delta K^\nu{}_\mu \right]$$

Ghost condensate/ghost inflation
Weakly broken Galileons
...

After restoring the Goldstone $K_{ij} \rightarrow K_{ij}[\pi]$

$$S[\pi_c] \supset \int d\eta d^3\mathbf{x} \left[\frac{a(\eta)}{\Lambda_\pi^2} \left(\gamma_2 \pi_c' (\nabla \pi_c)^2 + \gamma_3 \pi_c'^3 + \frac{\bar{\gamma}_5}{\Lambda_\pi^2} \frac{\pi_c'}{a^2} (\nabla^2 \pi_c)^2 + \frac{\bar{\gamma}_6}{\Lambda_\pi^2} \frac{\pi_c'}{a^2} (\partial_{ij} \pi_c)^2 \right) \right. \\ \left. \frac{1}{a^2(\eta) \Lambda_\pi^6} \left(\bar{\gamma}_{55} (\nabla \pi_c)^2 (\nabla^2 \pi_c)^2 + \bar{\gamma}_{66} (\nabla \pi_c)^2 (\partial_{ij} \pi_c)^2 + \bar{\gamma}_{11} \pi_c'^2 (\nabla^2 \pi_c)^2 + \bar{\gamma}_{12} \pi_c'^2 (\partial_{ij} \pi_c)^2 \right) \right]$$

$\pi \rightarrow \pi + \alpha$
 $\pi \rightarrow -\pi \quad t \rightarrow -t$

Many redundant operators can be removed through *disformal transformations* [Bordin et. al. (2017)]

Positivity constraints on the EFT of inflation

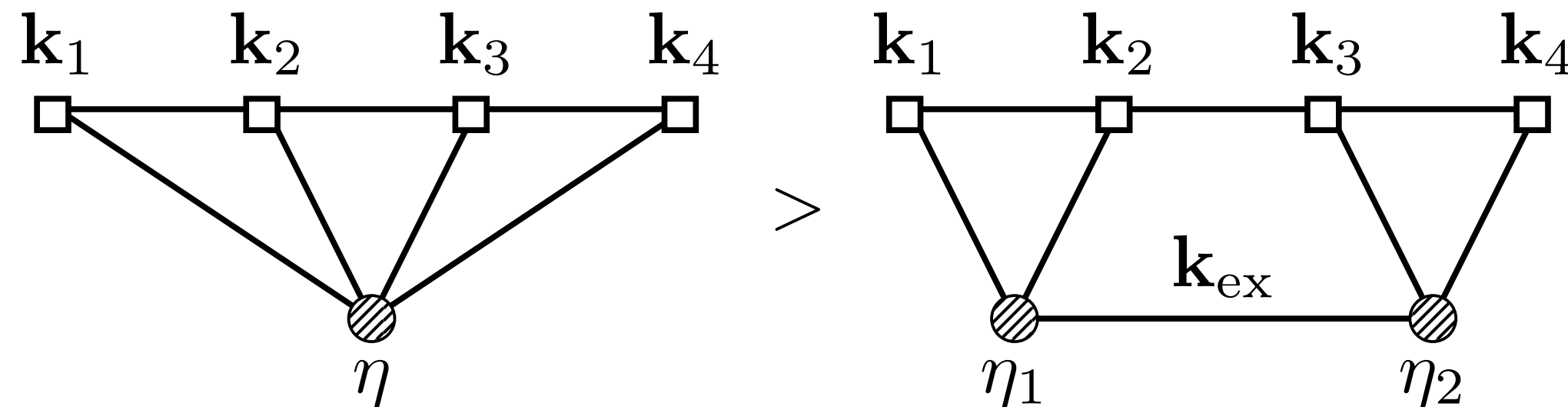
The extracted flat amplitude in the forward limit

$$\mathcal{A}(s, t \rightarrow 0) = \left[c_s^2 (\bar{\gamma}_{12} + \bar{\gamma}_{11}) - \bar{\gamma}_{66} - \bar{\gamma}_{55} + (\bar{\gamma}_5 + \bar{\gamma}_6)(3c_s^2 \gamma_3 + \gamma_2) \right] \frac{s^3}{8\Lambda_\pi^6 c_s^6}.$$

By applying the *negativity* condition $\mathcal{A}'''(s) \big|_{s=0} < 0$

$$\bar{\gamma}_{66} + \bar{\gamma}_{55} - c_s^2 (\bar{\gamma}_{12} + \bar{\gamma}_{11}) > (\bar{\gamma}_5 + \bar{\gamma}_6)(3c_s^2 \gamma_3 + \gamma_2)$$

Also satisfy the “*Suyama-Yamaguchi*” like structure!



Summary

- Evidence for new positivity bounds on the Wilson coefficients of the EFTol.
- Our approach relies in two ideas: the flat *space limit of cosmological correlators* and the analytic structure of a *superfluid*.
- Is the condition $(-1)^n \partial_s^n \mathcal{A}(s) \big|_{s=0} > 0$ a general feature of spontaneously broken Lorentz theories?
- Can we say something about specific models of inflation that don't satisfy the bounds?